Question 1:

Create a new directory called mysite in your home folder, place an index.html file inside it, and set permissions so only your user can read and access the site files.

Answer:

- **Step 1: Create the directory in your home folder**

mkdir -p ~/mysite

- **Step 2: Create the index.html file inside the directory**

echo "<h1>Welcome to My Private Site</h1>" > ~/mysite/index.html

- **Step 3: Restrict permissions to owner only**
    - Directory (read, write, execute for owner only):

chmod 700 ~/mysite

    - File (read and write for owner only):

chmod 600 ~/mysite/index.html

- **Step 4: Verify permissions**

ls -ld ~/mysite

(Output shows drwx------, confirming all group and other access is completely disabled).

Question 2:

Configure your Apache or Nginx server to serve the mysite directory as a website, then test if you can access it from your browser while logged in as your user.

Answer:

- **Step 1: Configure Apache Virtual Host / Alias**

Edit the Apache configuration file:

sudo nano /etc/apache2/sites-available/000-default.conf

Add the following block inside <VirtualHost *:80>:

Apache

Alias /mysite /home/<your_username>/mysite

<Directory /home/<your_username>/mysite>

Options Indexes FollowSymLinks

AllowOverride None

Require all granted

</Directory>

- **Step 2: Grant the web server access across the home directory hierarchy**

(Note: The web server runs as www-data. For it to traverse and serve files, the execute bit on your home path and read access for www-data must be permitted, or an ACL must be applied):

chmod o+x /home/<your_username>

setfacl -m u:www-data:rx ~/mysite

setfacl -m u:www-data:r ~/mysite/index.html

- **Step 3: Restart Apache**

sudo systemctl restart apache2

- **Step 4: Verification**

Open a browser and navigate to http://localhost/mysite. The page loads and displays "Welcome to My Private Site".

Question 3:

Change the permissions of the mysite directory so that another user on the system cannot access the site files directly. Try to access the files as the other user and note the result.

Hint: Use chmod and su or a different terminal window for the second user.

Answer:

- **Step 1: Create a test user (if not existing)**

sudo useradd -m -s /bin/bash guestuser

sudo passwd guestuser

- **Step 2: Restrict permissions on mysite directly**

chmod 750 ~/mysite

(Owner has full access, owner group has read/execute, and other users have zero permissions ---).

- **Step 3: Test access by switching to the second user**

su - guestuser

Attempt to list the directory:

ls -la /home/<your_username>/mysite

Attempt to read the file directly:

cat /home/<your_username>/mysite/index.html

- **Result & Observation:**

Both commands fail with the terminal error:

bash: cd: /home/<your_username>/mysite: Permission denied or cat: /home/<your_username>/mysite/index.html: Permission denied.

Because the "others" bit is 0 (---), the Linux kernel blocks guestuser from traversing the directory or reading its contents.

Question 4:

Document the steps you took to restrict access to your hosted site, including the commands used, permission settings, and screenshots or outputs showing successful and failed access attempts.

Answer:

- **1. Initial Setup Commands:**
    - mkdir -p ~/mysite
    - echo "<h1>Welcome to My Private Site</h1>" > ~/mysite/index.html
- **2. Web Server Configuration:**
    - Added Alias /mysite /home/<your_username>/mysite to /etc/apache2/sites-available/000-default.conf.
    - Reloaded daemon: sudo systemctl restart apache2.
- **3. Permission Hardening Commands:**
    - chmod 750 /home/<your_username>/mysite

- chmod 640 /home/<your_username>/mysite/index.html
- setfacl -m u:www-data:rx /home/<your_username>/mysite
- setfacl -m u:www-data:r /home/<your_username>/mysite/index.html

- **4. Terminal Outputs:**
  - **Permission verification output (ls -ld ~/mysite):**

drwxr-x---+ 2 <your_username> <your_username> 4096 Sep 16 21:00 /home/<your_username>/mysite

  - **Successful Browser Access:**

Navigating to http://localhost/mysite returns HTTP 200 OK rendering Welcome to My Private Site.

  - **Failed Direct User Access Output:**

Bash

$ su - guestuser

$ cat /home/<your_username>/mysite/index.html

cat: /home/<your_username>/mysite/index.html: Permission denied